Wednesday, 7 January 2026 6:05 PM

Step-by-step process to create a python project

- Install the python.
- Create a virtual environment → Isolate project dependencies.
  - ↳ python -m venv env/Day.
  - Activate → virtual environment
    - ↳ name_venv\Scripts\activate
- Installation of all the project dependencies. (flask or all the necessary libraries).
- Create a python project structure.

  Typical structure

  project name/
  - ↳ app.py.
  - ↳ venv/
  - ↳ templates.

- Write the python application.
  - Import.
  - Create a flask object
  - add routes / endpoints
  - run the application
- Run & test project
  - ↳ local host or loopback address (local host).
    - 127.0.0.1:5000
      - ↳ default flask port.
- Stop the application

  Pressing Ctrl+c.

Que Can your friend access your application from their laptop?

→ No

127.0.0.1:5000 → localhost

↳ accessible only from my computer

- How can other access your application
  - We have to deploy the application.

Deployment → our application will be on a publically reachable server.

- Can we deploy from our own system. → Yes (but it's not recommended).

Challenges of self hosting (local system)

- Hardware cost & scaling.

- Maintenance & monitoring cost.
- Electricity requirement. — 24x7
- Internet dependency.

Solution : Use cloud computing

Amazon AWS
Google Cloud Platform
microsoft azure
} Virtual infrastructure
↳ no physical machine that we need to manage.

- What do cloud provider actually offer.
  → Services
  ↳ 3 primary model.
  - SaaS. (Software as a Service)
    - Ready to use application
    - no deployment or infrastructure management

    Eg. Google docs,
    google drive
    google colab

    Responsibility 'Cloud provider handle everything.
  - PaaS (Platform as a Service)
    - You deploy code.
    - Platform manage OS., runtime, scaling.

    Eg. Flask/Spring. appx.
    APIs
    web applications
    } eg. AWS Elastic Beanstalk.
    - Responsibility split — Cloud → Infra + platform
      You → application code
  - IaaS (Infrastructure as a Service)
    - We get a virtual machine
    - Full control over OS & software

    Eg. Custom services
    advanced configuration
    Production system.

    eg. AWS EC2.

    Responsibility split

    Cloud → hardware, networking
    You → OS, security, application, scaling.

- Step by Step process : Deploying a Python application on AWS (IaaS model)
  - Step-1 : Create a AWS account
  - Step-2 rent a Server (virtual computer)
    ↳ launch an EC2 instance. → renting a computer
    - You choose RAM

    A virtual machine that include

- CPU
- RAM
- Storage (disk)
- Operating system (Linux / windows).

CPU
Storage.
OS.
- AWS provide you IP address.

we only select the configuration → AWS manages the physical hardware.

- **Step-3** : Transfer project from local system to AWS server.

Once the server is running, your code is still on your laptop so we must copy the project files to the AWS machine.

Tool → SCP (Secure Copy protocol).
↳ It work on SSH.

Step-4 : Remotely control the cloud server.

To configure & run the application, we must log into the server remotely.

Tool used – SSH (Secure shell).
↳ allow us to

- access the server terminal
- Install software
- Run command
- Manage files

Step-5 : Configure the server.

- Install python
- Install project dependencies
- Run the application